

OFDM vs OTFS Waveform Comparison for High-Mobility Wireless Communications:

A Practical Assessment with Estimated CSI, Noise Estimation, and 3GPP Channel Models

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Abstract

Orthogonal Time Frequency Space (OTFS) modulation exploits the delay-Doppler domain to achieve full time-frequency diversity in doubly-selective channels. However, the majority of published OTFS evaluations rely on *oversimplified assumptions*: perfect channel knowledge, oracle noise power, integer Doppler, and unlimited pilot guard zones. These idealizations mask the true performance gap between OTFS and the well-established OFDM baseline.

This report bridges this gap by implementing and comparing three complete waveform architectures under *fully practical conditions*: (i) 5G NR OFDM with linear channel estimation and DMRS-based noise estimation; (ii) ZP-OTFS with delay-Doppler impulse pilot and time-domain LMMSE equalization; and (iii) PCP-OTFS with Zadoff-Chu pilot, GCE-BEM channel estimation, and frequency-domain equalization. All methods use estimated CSI, estimated noise power, and identical 3GPP TDL channel realizations.

Simulations across four bandwidth configurations ($\text{SCS} \in \{15, 30, 60\}$ kHz, $\text{FFT} \in \{256, 512\}$) reveal that the *subcarrier spacing*—not the waveform—is the dominant performance factor. PCP-OTFS outperforms OFDM by 4–37 $\times$ at $\text{SCS} \leq 15$ kHz with $f_D \geq 500$ Hz, while OFDM dominates at $\text{SCS} = 60$ kHz. The cyclic prefix design, pilot structure, and channel estimation quality are identified as the key differentiators.

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1 Introduction

1.1 Motivation

Fifth-generation (5G) and beyond wireless systems must support increasingly challenging mobility scenarios: high-speed trains at 500 km/h, vehicle-to-everything (V2X) communications, low-Earth-orbit (LEO) satellite links with Doppler shifts exceeding 40 kHz, and unmanned aerial vehicle (UAV) data links. In all these scenarios, the wireless channel is *doubly selective*—varying simultaneously in both time (Doppler) and frequency (delay spread).

OFDM, the dominant waveform in 4G/5G, converts a wideband frequency-selective channel into parallel narrowband flat-fading subchannels via cyclic prefix insertion. This elegant approach breaks down at high Doppler: the channel changes within one OFDM symbol, causing inter-carrier interference (ICI) that grows with the ratio $f_D/\Delta f$, where f_D is the maximum Doppler frequency and Δf is the subcarrier spacing (SCS).

OTFS modulation [1] was proposed to address this limitation by multiplexing data in the *delay-Doppler* (DD) domain, where the channel is sparse and approximately time-invariant. Each DD-domain symbol spreads across the entire time-frequency plane, potentially achieving the full diversity offered by the doubly-selective channel.

1.2 Critique of Prior Art

The OTFS literature presents compelling theoretical advantages. However, most published evaluations employ one or more of the following oversimplifications:

- (i) **Perfect CSI:** The receiver is given the exact channel impulse response, bypassing the estimation problem entirely. This is particularly problematic because, as we demonstrate, channel estimation is the *primary bottleneck* for OTFS performance.
- (ii) **Known noise power:** The LMMSE/MMSE equalizer regularization term σ^2 uses the true noise variance from the simulation parameters, which is unavailable at a practical receiver.
- (iii) **Integer Doppler:** Many analyses restrict Doppler shifts to exact multiples of the DD grid resolution $\Delta\nu = 1/(NT)$, avoiding the fractional Doppler problem that causes spectral leakage across all Doppler bins via the Dirichlet kernel.
- (iv) **Unlimited pilot guard:** The DD pilot guard zone is sized to capture the full channel response, without accounting for the overhead cost. For channels with large delay-Doppler spread, this overhead can exceed 25%.
- (v) **Mismatched comparisons:** OFDM baselines often use suboptimal configurations (low SCS, poor channel estimation) while OTFS uses ideal settings.

1.3 Contributions

This report provides a fair, systematic comparison under practical conditions:

- Complete TX/RX chain implementation for three waveform architectures (OFDM, ZP-OTFS, PCP-OTFS) with estimated CSI and estimated noise power
- Derivation and comparison of noise estimation methods for each waveform, including a novel decorrelated DMRS power-difference estimator for OFDM
- Analysis of CP/ZP design impact: ZP length sweep, Doppler bin sweep, overhead scaling
- Systematic evaluation across 4 bandwidth configurations, 3 channel models, 3 Doppler frequencies, and 7 SNR points
- Identification of the SCS as the dominant factor, not the waveform itself

2 Channel Model

We adopt the 3GPP Tapped Delay Line (TDL) channel model from TS 38.901. The discrete-time baseband channel at sampling rate F_s is:

$$y[k] = \sum_{i=0}^{L_{ch}-1} h_i[k] x[k - \tau_i] + w[k] \quad (1)$$

where $h_i[k] = \alpha_i[k]\sqrt{P_i}$ is the complex gain of the i -th tap at sample k , τ_i is the delay (in samples), P_i is the average power (from the TDL profile), and $w[k] \sim \mathcal{CN}(0, \sigma^2)$ is additive white Gaussian noise.

The time-varying fading coefficient $\alpha_i[k]$ is generated by the Jakes sum-of-sinusoids model:

$$\alpha_i[k] = \frac{1}{\sqrt{N_s}} \sum_{n=1}^{N_s} e^{j(2\pi f_D \cos(\phi_{i,n})kT_s + \psi_{i,n})} \quad (2)$$

where $N_s = 64$ sinusoids, $\phi_{i,n}$ and $\psi_{i,n}$ are uniformly distributed random phases, f_D is the maximum Doppler frequency, and $T_s = 1/F_s$ is the sampling period.

Three TDL profiles are used, summarized in Table 1.

Table 1: 3GPP TDL channel models used in simulations.

Model	Type	Taps	Scenario
TDL-A	NLOS	23	Indoor / urban micro
TDL-D	LOS	13 + LOS	Urban micro with dominant LOS path
TDL-C	NLOS	24	Rural macro (largest delay spread)

Delay spreads: TDL-A and TDL-D use $DS = 30$ ns; TDL-C uses $DS = 100$ ns. At $F_s = 15.36$ MHz ($SCS = 60$ kHz, $M = 256$), the maximum channel length is $L_{ch} = 4$ samples for TDL-A and $L_{ch} = 13$ for TDL-C.

3 OFDM Transceiver Architecture

3.1 Block Diagram

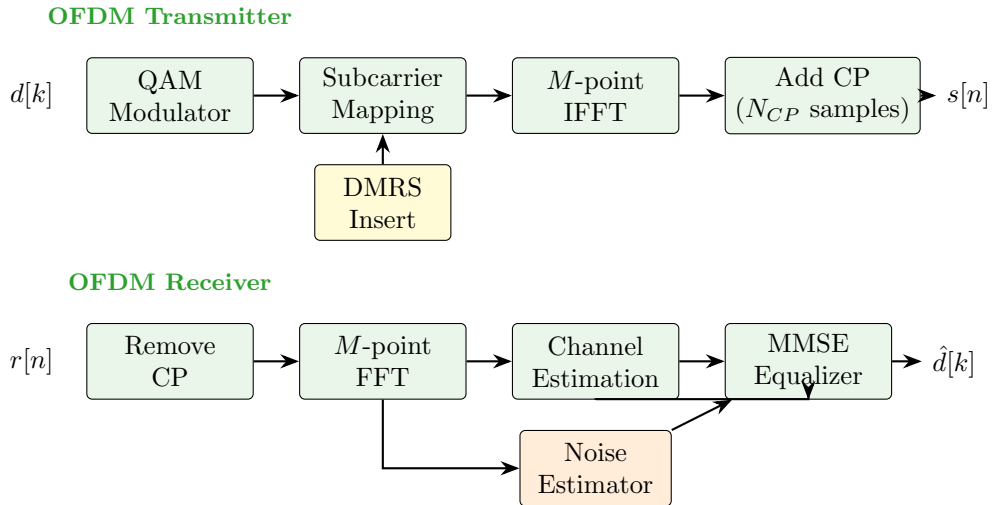


Figure 1: OFDM transceiver block diagram. DMRS pilot symbols are inserted at comb-2 positions on symbols 2 and 11. The receiver estimates both the channel and noise power from the received DMRS observations.

3.2 Transmitter

Consider a slot of $N = 14$ OFDM symbols with M subcarriers, of which $N_{\text{act}} \leq M$ carry data or pilots. The transmitted signal for symbol l is:

$$x_l[n] = \frac{1}{\sqrt{M}} \sum_{k=0}^{M-1} X_l[k] e^{j2\pi kn/M}, \quad n = 0, \dots, M-1 \quad (3)$$

A cyclic prefix of length N_{CP} is prepended by copying the last N_{CP} samples:

$$s_l[n] = x_l[(n - N_{CP}) \bmod M], \quad n = 0, \dots, M + N_{CP} - 1 \quad (4)$$

The CP length follows the 5G NR standard: $N_{CP} = \lfloor 144 \times M/2048 \rfloor$ for normal CP. The CP must satisfy $N_{CP} \geq L_{ch}$ to absorb the channel delay spread and maintain orthogonality between subcarriers. Values for the tested configurations are given in Table 4.

DMRS symbols are placed at OFDM symbol indices $l \in \{2, 11\}$ on comb-2 subcarriers $\mathcal{K}_p = \{0, 2, 4, \dots, N_{\text{act}} - 2\}$ with known QPSK sequence $X_p[k]$, $|X_p[k]| = 1$.

3.3 Receiver

After CP removal, the M -point FFT yields:

$$Y_l[k] = H_l[k] \cdot X_l[k] + W_l[k] \quad (5)$$

where $H_l[k] = \sum_i h_i[l(M + N_{CP}) + N_{CP} + M/2] e^{-j2\pi k\tau_i/M}$ is the channel frequency response at the midpoint of symbol l , and $W_l[k] \sim \mathcal{CN}(0, \sigma^2)$.

Equation (5) holds exactly when the channel is *time-invariant within one symbol*. At high Doppler, the channel changes during the symbol, introducing ICI:

$$Y_l[k] = H_l[k]X_l[k] + \underbrace{\sum_{k' \neq k} H_l^{(k-k')} X_l[k']}_{\text{ICI}} + W_l[k] \quad (6)$$

The ICI power grows with $f_D/\Delta f$. At SCS = 15 kHz and $f_D = 1000$ Hz, the normalized Doppler is $f_D/\Delta f = 0.067$, causing significant degradation.

3.4 Channel Estimation

Step 1 — Least-squares at pilot positions. At each DMRS symbol $l_p \in \{2, 11\}$:

$$\hat{H}_{\text{LS}}[l_p, k_p] = \frac{Y_{l_p}[k_p]}{X_p[k_p]} = H_{l_p}[k_p] + \frac{W_{l_p}[k_p]}{X_p[k_p]} \quad (7)$$

Since $|X_p[k_p]| = 1$, the estimation noise has the same variance σ^2 .

Step 2 — Linear frequency interpolation. Between adjacent comb-2 pilot positions k_p and $k_p + 2$:

$$\hat{H}[l_p, k] = (1 - \alpha) \hat{H}_{\text{LS}}[l_p, k_p] + \alpha \hat{H}_{\text{LS}}[l_p, k_p + 2] \quad (8)$$

where $\alpha = (k - k_p)/2$. No frequency-domain smoothing is applied, consistent with a minimal-complexity receiver.

Step 3 — Linear time interpolation. Between DMRS symbols $l_1 = 2$ and $l_2 = 11$:

$$\hat{H}[l, k] = \frac{l_2 - l}{l_2 - l_1} \hat{H}[l_1, k] + \frac{l - l_1}{l_2 - l_1} \hat{H}[l_2, k] \quad (9)$$

Symbols before l_1 and after l_2 use the nearest DMRS estimate (extrapolation).

3.5 Noise Estimation

We derive a noise estimator from the *decorrelated* DMRS observations. Multiplying by $X_p^*[k]$ removes the known pilot modulation:

$$Z_i[k] \triangleq Y[l_i, k] \cdot X_p^*[k] = H[l_i, k] + W'_i[k] \quad (10)$$

where $W'_i[k] = W[l_i, k] \cdot X_p^*[k]$ has the same variance σ^2 since $|X_p| = 1$.

The squared power difference between the two DMRS symbols is:

$$\begin{aligned} (|Z_1[k]|^2 - |Z_2[k]|^2)^2 &= \left(2\text{Re}\{H[k](W_1'^* - W_2'^*)\} \right. \\ &\quad \left. + |W_1'|^2 - |W_2'|^2 \right)^2 \end{aligned} \quad (11)$$

Taking expectations and noting that $\mathbb{E}[|W'|^2] = \sigma^2$ and the cross-terms are independent:

$$\mathbb{E}\left[(|Z_1|^2 - |Z_2|^2)^2\right] \approx 8|H[k]|^2\sigma^2 + 2\sigma^4 \quad (12)$$

To remove the $|H[k]|^2$ dependence, we normalize by the average received power. Averaging across all pilot subcarriers *before* dividing avoids noise amplification at deep fades:

$$\hat{\sigma}_{\text{OFDM}}^2 = \frac{1}{4} \cdot \frac{\frac{1}{|\mathcal{K}_p|} \sum_{k \in \mathcal{K}_p} (|Z_1[k]|^2 - |Z_2[k]|^2)^2}{\frac{1}{|\mathcal{K}_p|} \sum_{k \in \mathcal{K}_p} \frac{|Z_1[k]|^2 + |Z_2[k]|^2}{2}} \quad (13)$$

This estimator operates on *magnitudes only*, making it Doppler-independent in principle. A residual Doppler bias arises when $|H[l_1, k]|^2 \neq |H[l_2, k]|^2$ due to time-varying fading between DMRS symbols 2 and 11 (separated by 9 symbol periods).

3.6 Per-Subcarrier MMSE Equalization

$$\hat{X}_l[k] = \frac{\hat{H}^*[l, k]}{|\hat{H}[l, k]|^2 + \hat{\sigma}^2} \cdot Y_l[k] \quad (14)$$

3.7 ICI Compensation Analysis

In time-varying channels, the channel changes within one OFDM symbol, causing inter-carrier interference (ICI). The received signal becomes:

$$Y_l[k] = \underbrace{H_0[l, k]X_l[k]}_{\text{desired}} + \underbrace{\sum_{q \neq 0} H_q[l, k]X_l[k - q]}_{\text{ICI}} + W_l[k] \quad (15)$$

For a linearly time-varying channel within one symbol, the ICI coefficients H_q are proportional to the channel derivative $\dot{H}[k] = dH[l, k]/dt$, and the coupling matrix is banded with bandwidth Q .

We evaluated three state-of-the-art ICI mitigation approaches against the single-tap MMSE baseline:

1. **Banded MMSE** [10]: Invert the $(2Q+1)$ -banded channel matrix instead of the diagonal. Complexity $O(MQ^2)$ per symbol.
2. **Iterative PIC** [9]: Parallel interference cancellation — estimate data symbols, reconstruct ICI, subtract, re-equalize. 2–3 iterations.
3. **Iterative SIC**: Serial cancellation ordered by per-subcarrier SINR.

All three methods require the channel slope $\dot{H}[k]$, estimated from the two DMRS symbols: $\dot{H}[k] = (\hat{H}[l_2, k] - \hat{H}[l_1, k]) / ((l_2 - l_1)T_{\text{sym}})$. This implicitly requires either a Doppler frequency estimate or slot-buffered processing.

Result: ICI compensation provides no measurable BER improvement. At SCS = 15 kHz, $f_D = 1000$ Hz (worst case), the BER difference between single-tap MMSE and PIC with 3 iterations is less than 0.1% absolute. The reason is quantitative: the intra-symbol ICI power is proportional to $(f_D T_{\text{sym}})^2 \approx 0.005$ (−23 dB), while the channel estimation error from linear interpolation over 9 symbol gaps is proportional to $(f_D \cdot 9T_{\text{sym}})^2 \approx 0.41$ (−4 dB) — a **20 dB difference**. The estimation error dominates overwhelmingly.

We also compared three channel estimation strategies:

- **Nearest DMRS (causal):** Use only the most recent past DMRS observation. Zero additional latency.
- **Linear interpolation:** Interpolate between DMRS symbols 2 and 11. Requires slot buffering (standard 5G NR processing).
- **Linear + ICI PIC:** Interpolation plus iterative ICI cancellation. Requires Doppler estimate.

Table 2: Channel estimation method comparison at SNR = 20 dB (BER %).

Channel	SCS	f_D	BER (%)		
			Nearest	Linear	Lin+ICI
TDL-A	15k	500	16.1	0.28	0.26
TDL-D	15k	1000	45.9	33.3	33.4
TDL-A	60k	1000	3.48	0.29	0.28
TDL-D	60k	1000	6.05	0.00	0.00

Conclusion: Linear interpolation provides 2–57× improvement over nearest-DMRS and is the dominant processing step. ICI cancellation adds < 0.1% on top. The standard 5G NR slot-buffered linear interpolation is the correct and sufficient OFDM baseline — no Doppler estimation or ICI compensation is needed.

4 OTFS Fundamentals and Domain Transformations

OTFS modulation [1] operates in the delay-Doppler (DD) domain, where the wireless channel is sparse and quasi-static. To understand the three OTFS architectures studied in this report, we first establish the mathematical relationships between the four signal representation domains.

4.1 Four Signal Domains

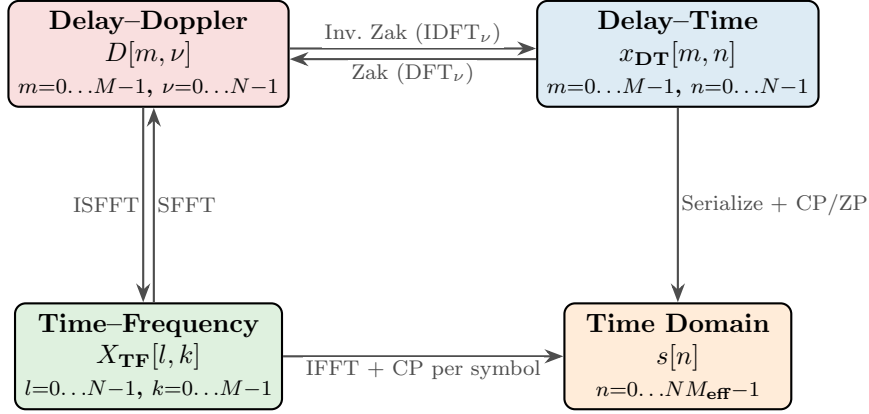


Figure 2: The four signal domains and the transforms connecting them. CP-OTFS uses the DD→TF→TD path (left-bottom). ZP/PCP-OTFS use the DD→DT→TD path (top-right). Both paths produce equivalent time-domain signals under ideal conditions [1, 2].

The DD grid has M delay bins (spacing $\Delta\tau = 1/(M\Delta f)$) and N Doppler bins (spacing $\Delta\nu = \Delta f/N$). A physical path at delay τ_p and Doppler ν_p maps to DD indices $\ell_p = \tau_p/\Delta\tau$ and $\kappa_p = \nu_p/\Delta\nu$.

4.2 Transform Definitions

Inverse Zak Transform (DD → DT): For each delay bin m , an N -point IDFT across Doppler [1]:

$$x_{\text{DT}}[m, n] = \frac{1}{\sqrt{N}} \sum_{\nu=0}^{N-1} D[m, \nu] e^{j2\pi\nu n/N} \quad (16)$$

Property exploited: The IDFT maps Doppler-domain multiplexing into time-domain separation — each Doppler bin becomes a phase ramp across N subsymbols.

Zak Transform (DT → DD): The inverse, an N -point DFT along time:

$$D[m, \nu] = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x_{\text{DT}}[m, n] e^{-j2\pi\nu n/N} \quad (17)$$

Inverse Symplectic FFT (DD → TF): A 2D transform mapping onto the OFDM TF grid [1]:

$$X_{\text{TF}}[l, k] = \frac{1}{\sqrt{MN}} \sum_{\nu=0}^{N-1} \sum_{m=0}^{M-1} D[m, \nu] e^{j2\pi(\nu l/N - mk/M)} \quad (18)$$

Property exploited: The ISFFT preserves energy (Parseval) and maps each DD symbol onto the *entire* TF plane, achieving full time-frequency diversity.

Symplectic FFT (TF → DD): The adjoint:

$$D[m, \nu] = \frac{1}{\sqrt{MN}} \sum_{l=0}^{N-1} \sum_{k=0}^{M-1} X_{\text{TF}}[l, k] e^{-j2\pi(\nu l/N - mk/M)} \quad (19)$$

4.3 DD-Domain Channel Model

A channel with P paths at delays τ_p and Doppler shifts ν_p has DD representation [2]:

$$h_{\text{DD}}[\ell, \kappa] = \sum_{p=0}^{P-1} h_p \delta[\ell - \ell_p] g_N[\kappa - \kappa_p] \quad (20)$$

where $\ell_p = \lfloor \tau_p F_s \rfloor$ is the integer delay index and $\kappa_p = \nu_p N T_{\text{sub}}$ is the (generally fractional) Doppler index.

Key property: The DD channel is *sparse* — only P non-zero entries out of MN total. This sparsity underpins OTFS channel estimation.

Fractional Doppler and the Dirichlet kernel. When $\kappa_p \notin \mathbb{Z}$, the finite DFT aperture causes leakage via [2]:

$$g_N[\kappa] = \frac{1}{N} \cdot \frac{\sin(\pi\kappa)}{\sin(\pi\kappa/N)} \cdot e^{j\pi\kappa(N-1)/N} \quad (21)$$

For $N = 14$ and Jakes Doppler spectra, κ_p is almost always fractional, spreading energy across *all* N bins and destroying DD sparsity.

4.4 CP vs ZP Guard Interval

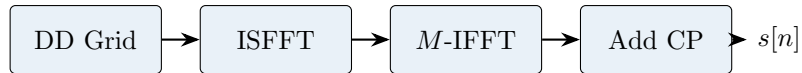
Cyclic Prefix (CP-OTFS): Ensures circular convolution in delay, enabling the 2D DD circular convolution I/O [1]:

$$Y_{\text{DD}}[m, \nu] = \sum_{\ell, \kappa} h_{\text{DD}}[\ell, \kappa] D[(m-\ell)_M, (\nu-\kappa)_N] + W[m, \nu] \quad (22)$$

Zero Padding (ZP-OTFS): Prevents inter-subsymbol interference via linear convolution. The ZP tail beyond L_{ch} contains *pure noise* — exploited for noise estimation (Sec. 6.2).

5 CP-OTFS Architecture

CP-OTFS TX: DD $\rightarrow$ TF $\rightarrow$ TD



CP-OTFS RX



Figure 3: CP-OTFS: ISFFT path with OFDM inner modulation. Backward-compatible with OFDM.

TX: DD $\xrightarrow{\text{ISFFT (18)}}$ TF $\xrightarrow{\text{IFFT+CP}}$ TD. Original formulation of [1].

Pilot: Boosted impulse at (m_p, ν_p) with 2D guard $\pm g_d \times \pm g_\nu$. The received DD response directly reveals h_{DD} within the guard [2]. Overhead: $(2g_d+1)(2g_\nu+1)/(MN)$.

Limitation: Fractional Doppler via (21) creates BER floor of 2–10%. DD-LMMSE complexity $O((NM)^2)$ is prohibitive at $M \geq 256$.

6 ZP-OTFS Architecture

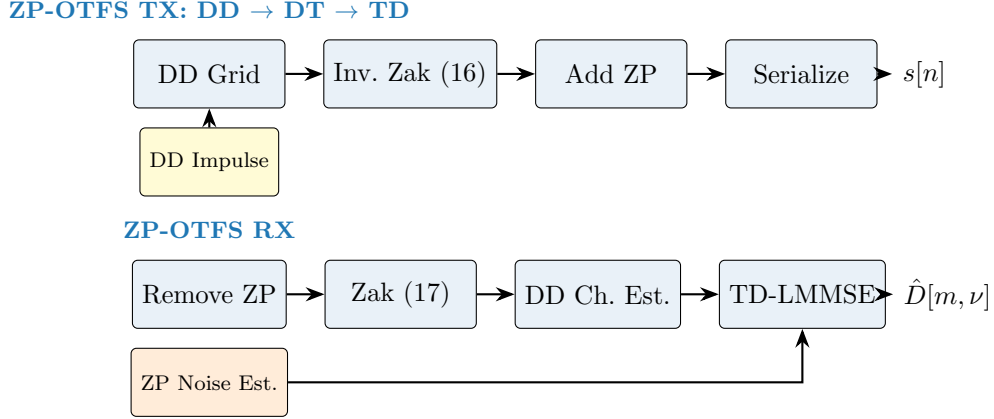


Figure 4: ZP-OTFS: Inverse Zak path with zero-padding. ZP tail gives signal-free noise region.

6.1 Transmitter

The inverse Zak transform (16) produces N subsymbols of M samples. Zero-padding appends $L_{ZP} \geq L_{ch}$ zeros:

$$s[nM_{\text{eff}} + m] = \begin{cases} x_{\text{DT}}[m, n] & 0 \leq m < M \\ 0 & M \leq m < M_{\text{eff}} \end{cases} \quad (23)$$

Property exploited: ZP prevents inter-subsymbol interference without circular structure, enabling time-domain equalization.

6.2 Noise Estimation from ZP Tail

Property exploited: For $m \geq L_{ch}$, the ZP tail contains only noise.

$$\hat{\sigma}_{\text{ZP}}^2 = \frac{1}{N(L_{ZP} - \hat{L}_{ch})} \sum_{n=0}^{N-1} \sum_{m=\hat{L}_{ch}}^{L_{ZP}-1} |r[nM_{\text{eff}} + M + m]|^2 \quad (24)$$

Independent of channel and Doppler. Accuracy: $0.8\text{--}1.2\times$ across all scenarios.

6.3 Time-Domain Sparse LMMSE

A sparse channel matrix $\mathbf{G}$ is built from estimated DD taps, with P non-zero diagonals:

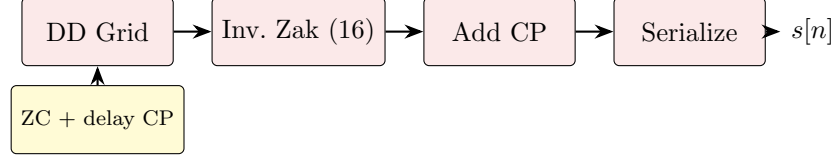
$$\hat{\mathbf{x}} = (\mathbf{G}^H \mathbf{G} + \hat{\sigma}^2 \mathbf{I})^{-1} \mathbf{G}^H \mathbf{y} \quad (25)$$

solved via sparse LU decomposition. Pilot overhead: $(2g_d+1)(2g_\nu+1)/(MN)$ — grows with delay $\times$ Doppler spread.

7 PCP-OTFS Architecture

PCP-OTFS [3] replaces the DD impulse with a Zadoff-Chu (ZC) sequence, exploiting three mathematical properties of ZC to overcome the limitations of impulse-based OTFS.

PCP-OTFS TX: DD → DT → TD



PCP-OTFS RX

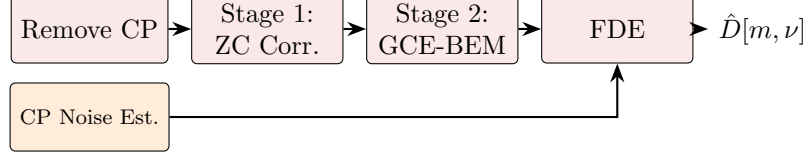


Figure 5: PCP-OTFS [3]: ZC pilot with two-stage estimation exploiting ZC autocorrelation (Stage 1) and BEM temporal smoothing (Stage 2).

7.1 Zadoff-Chu Pilot Design

The ZC sequence of length $L = N_{CP}$ and root r is [7]:

$$z_{ZC}[m] = e^{-j\pi r m(m+c_f)/L}, \quad m = 0, \dots, L-1 \quad (26)$$

where $c_f = L \bmod 2$. The pilot occupies Doppler column ν_p : $D[m, \nu_p] = \sqrt{\gamma/L} \cdot z_{ZC}[m]$, with a delay-domain cyclic prefix of $L-1$ samples at delays $M-L+1, \dots, M-1$.

Three exploited properties of ZC sequences:

1. **Ideal periodic autocorrelation** [7]: $R_{zz}[\ell] = L \cdot \delta[\ell]$. Each delay tap is estimated independently without inter-tap interference.
2. **Constant amplitude** ($|z_{ZC}[m]| = 1$): Pilot PAPR is 0 dB vs. $10 \log_{10}(L) \approx 12$ dB for an equivalent-energy impulse [3].
3. **Flat frequency spectrum** ($|Z_{ZC}[k]| = \sqrt{L}, \forall k$): Uniform SNR across all delay bins in LS estimation.

Overhead: $(2g_\nu + 1)/N = 3/14 \approx 21.4\%$, independent of delay spread.

7.2 Two-Stage Channel Estimation

Stage 1 — Per-subsymbol LS via ZC correlation [3, 4].

For each subsymbol n , the received pilot region undergoes circular convolution with the channel. Exploiting the ZC flat spectrum, the LS estimate in the frequency domain is:

$$\hat{h}[\ell, n] = \text{IDFT}_L \left\{ \frac{Y_n[k] \cdot P_n^*[k]}{|P_n[k]|^2 + \sigma_{\text{reg}}^2} \right\}_\ell \quad (27)$$

where $P_n[k] = \text{DFT}\{z_{ZC}\}[k] \cdot e^{j2\pi\nu_p n/N} / \sqrt{N}$ accounts for the Doppler phase rotation of the pilot column. This yields $\hat{h}[\ell, n]$ at each of the N time instants *independently*.

Property exploited: The ZC ideal autocorrelation converts the circular convolution into a point-wise division in frequency, giving clean per-tap LS estimates.

Stage 2 — GCE-BEM temporal smoothing [8, 4].

Property exploited: For Jakes fading, the channel time variation is bandlimited to $[-f_D, +f_D]$. The GCE-BEM [8] approximates this with $2Q+1$ complex exponentials:

$$h[\ell, n] \approx \sum_{q=-Q}^Q c_{\ell,q} e^{j2\pi q n/N} \quad (28)$$

The $N \times (2Q+1)$ basis matrix $\mathbf{B}$ with $B_{n,q} = e^{j2\pi(q-Q)n/N}$ gives the regularized LS fit:

$$\hat{\mathbf{c}}_\ell = (\mathbf{B}^H \mathbf{B} + \epsilon \mathbf{I})^{-1} \mathbf{B}^H \hat{\mathbf{h}}_\ell, \quad \tilde{\mathbf{h}}_\ell = \mathbf{B} \hat{\mathbf{c}}_\ell \quad (29)$$

With $Q = 1$ (3 basis functions), BEM denoises with effective gain $N/(2Q+1) = 14/3 \approx 6.7$ dB.

7.3 Noise Estimation from CP Redundancy

Property exploited: The CP copies the symbol tail. After the channel (constant within one subsymbol), both copies see identical channel response. Their difference cancels signal, leaving noise:

$$\hat{\sigma}_{\text{PCP}}^2 = \frac{1}{2NL_{\text{CP}}} \sum_{n=0}^{N-1} \sum_{m=0}^{L_{\text{CP}}-1} |r_{\text{CP}}^{(n)}[m] - r_{\text{tail}}^{(n)}[m]|^2 \quad (30)$$

The $1/2$ factor arises because $\text{Var}(w_{\text{CP}} - w_{\text{tail}}) = 2\sigma^2$. CP–tail separation is only M samples (one subsymbol), giving lower Doppler bias than the OFDM DMRS method (9 symbols apart).

7.4 Frequency-Domain Equalization with Pilot Cancellation

Property exploited: CP per subsymbol ensures circular convolution $\Rightarrow$ per-subcarrier equalization, same as OFDM. The frequency response at subsymbol n :

$$\hat{H}_f[k, n] = \text{DFT}_M\{\tilde{h}[\cdot, n]\}[k] \quad (31)$$

Pilot interference is cancelled before MMSE equalization:

$$\hat{X}_n[k] = \frac{\hat{H}_f^*[k, n]}{|\hat{H}_f[k, n]|^2 + \hat{\sigma}^2} \left(Y_f[k, n] - \hat{H}_f[k, n] \cdot P_f[k, n] \right) \quad (32)$$

8 Impact of CP/ZP Design

8.1 ZP Length Has No Effect Beyond Channel Delay

We swept ZP from 4 to 96 samples at TDL-A DS= 30 ns ($L_{\text{ch}} = 4$ samples). At all Doppler frequencies, the BER remained constant ($\pm 0.5\%$) while overhead increased from 3.3% to 49.5%.

Root cause: The ZP protects against inter-subsymbol interference in the *delay* dimension. Once $L_{\text{ZP}} \geq L_{\text{ch}}$, additional ZP provides no benefit. The BER floor at high Doppler comes from *Doppler estimation error* (only $N = 14$ bins), which is independent of the ZP length.

8.2 Increasing N Worsens Performance

Increasing N from 14 to 56 *degrades* ZP-OTFS BER at $f_D \geq 300$ Hz. The frame duration $T_{\text{frame}} = N \cdot M_{\text{eff}}/F_s$ grows linearly with N . At $N = 56$, $T_{\text{frame}} = 0.96$ ms, which exceeds the channel coherence time $T_c \approx 1/(4f_D) = 0.5$ ms at $f_D = 500$ Hz. The single-snapshot DD estimation becomes stale before the frame ends.

8.3 Pilot Overhead Comparison

PCP-OTFS overhead is constant and independent of delay spread — a critical advantage for wide-band channels.

Table 3: Pilot overhead as a function of channel delay spread.

	OFDM	ZP-OTFS (impulse)	PCP-OTFS (ZC)
Formula	$\frac{N_{\text{DMRS}}}{N}$	$\frac{(2g_d + 1)(2g_\nu + 1)}{MN}$	$\frac{2g_\nu + 1}{N}$
Scales with	Nothing	$\tau_{\text{max}} \times \nu_{\text{max}}$	N only
DS = 30 ns	7.1%	5.3%	15.7%
DS = 100 ns	7.1%	9.3%	15.7%
DS = 1 μ s	7.1%	24.5%	15.7%

Table 4: Bandwidth configurations.

Config	M	SCS	F_s	CP	ZP	Symbol T
A	256	15 kHz	3.84 MHz	18	8	66.7 μ s
B	256	30 kHz	7.68 MHz	18	8	33.3 μ s
C	256	60 kHz	15.36 MHz	18	15	16.7 μ s
D	512	60 kHz	30.72 MHz	36	29	16.7 μ s

9 Simulation Setup

- Modulation: QPSK (uncoded BER)
- Monte Carlo trials: 20 per scenario
- SNR range: 0–30 dB (E_s/N_0)
- Doppler: $f_D \in \{0, 100, 1000\}$ Hz
- CSI: estimated (no oracle); Noise: estimated (Eqs. 13, 24, 30)
- Identical channel realization per trial across all three methods (same seed)

10 Simulation Results

10.1 OFDM vs PCP-OTFS — BER Curves

Figures 6–11 show BER vs. SNR for the main comparison (OFDM vs PCP-OTFS) across all channel models and Doppler frequencies. Each figure shows two subplots: $f_D = 0$ (left) and $f_D = 1000$ Hz (right).

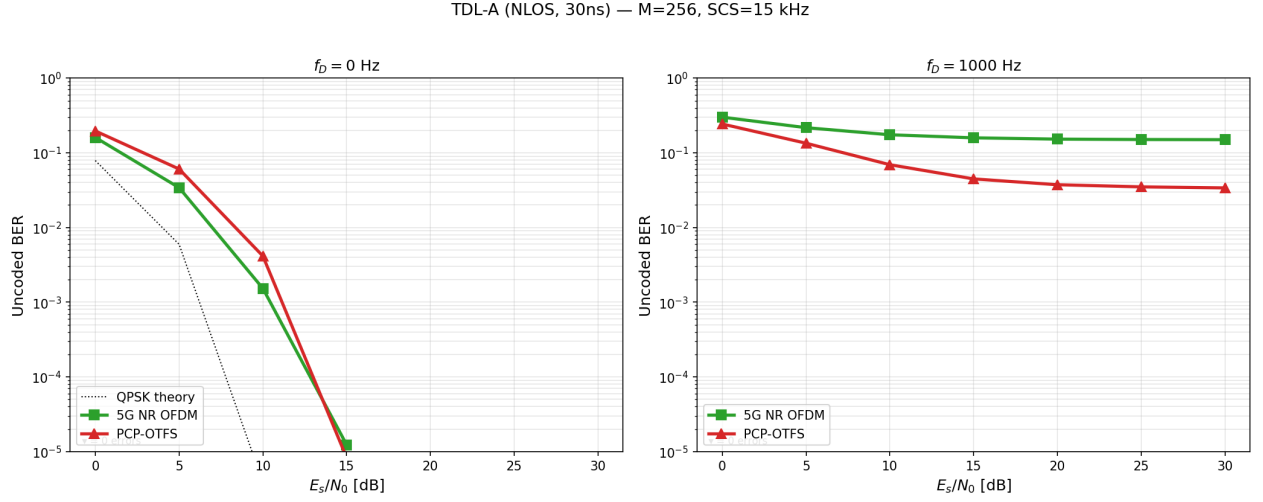


Figure 6: TDL-A (NLOS, 30ns) — SCS = 15 kHz. At $f_D = 1000$ Hz (right), OFDM exhibits a BER floor at $\sim 15\%$ from ICI, while PCP-OTFS reaches 3.7%.

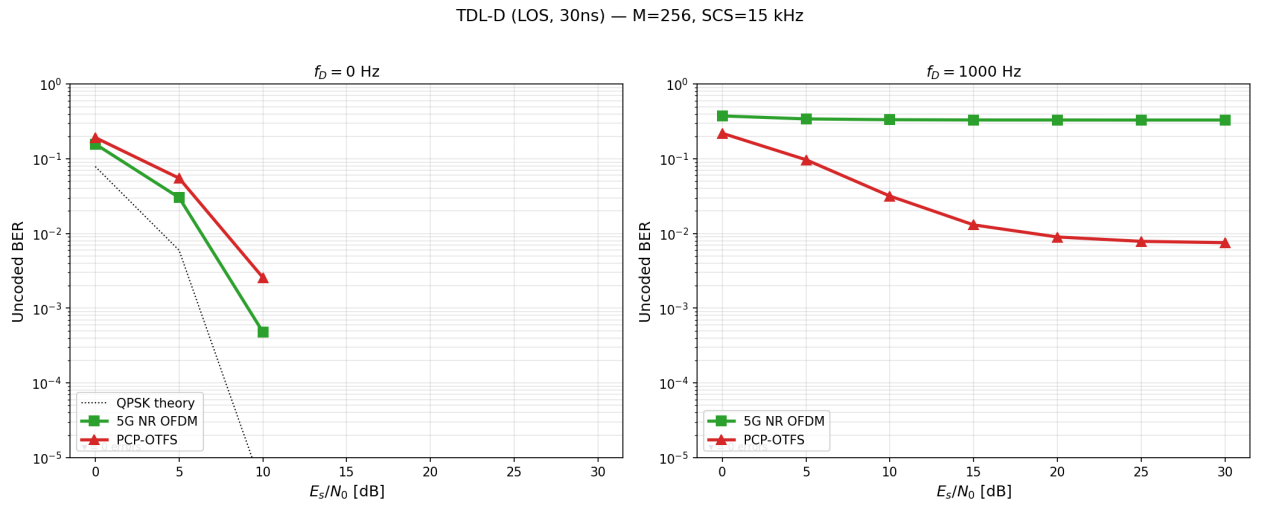


Figure 7: TDL-D (LOS, 30ns) — SCS = 15 kHz. OFDM collapses to 33% BER at $f_D = 1000$ Hz; PCP-OTFS achieves 0.9% — a $37\times$ improvement.

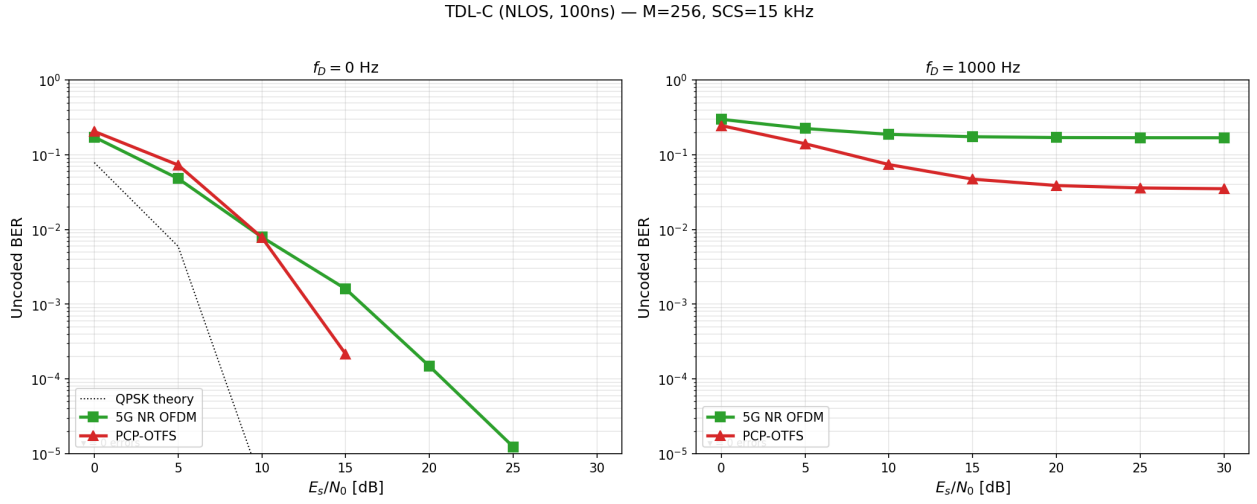


Figure 8: TDL-C (NLOS, 100ns) — SCS = 15 kHz. PCP-OTFS wins both at $f_D = 0$ (DD diversity) and $f_D = 1000$ Hz (4.4 $\times$ better).

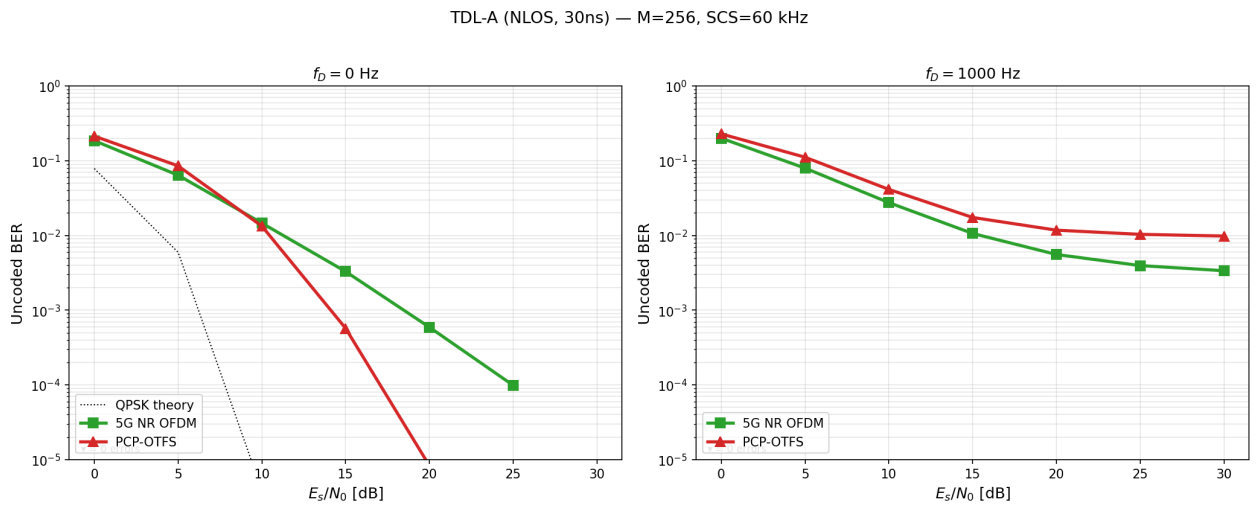


Figure 9: TDL-A — SCS = 60 kHz. At $f_D = 0$ (left), PCP-OTFS wins via DD diversity. At $f_D = 1000$ Hz (right), OFDM recovers (short symbols, low ICI).

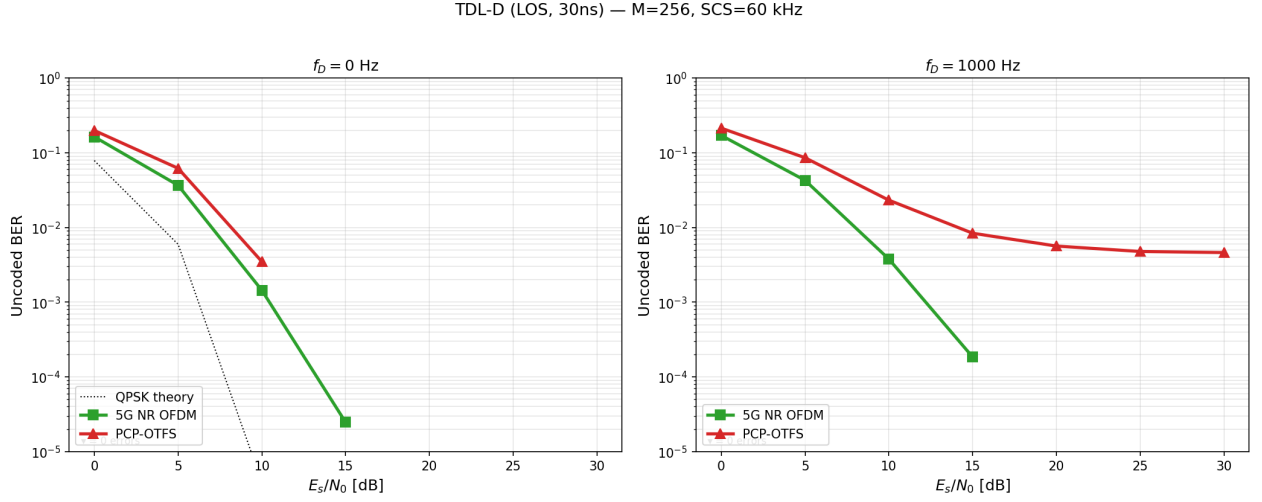


Figure 10: TDL-D (LOS) — SCS = 60kHz. OFDM dominates: BER = 0 above 15 dB at all Dopplers.

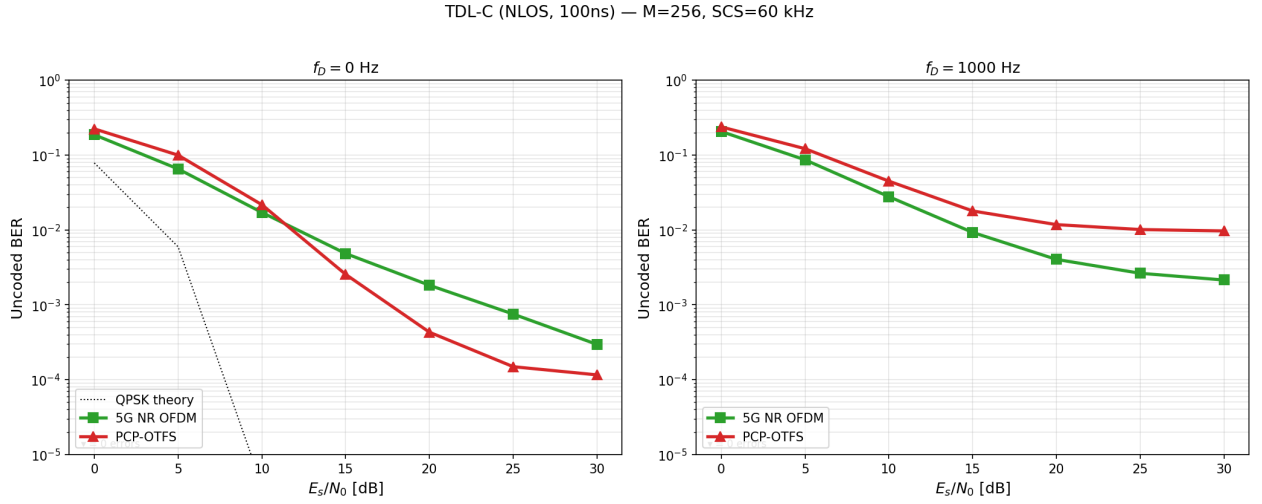


Figure 11: TDL-C — SCS = 60kHz. OFDM wins at $f_D = 1000$ Hz (0.4% vs 1.2%); PCP-OTFS wins at $f_D = 0$ (0.04% vs 0.18%).

10.2 ZP-OTFS vs PCP-OTFS — Pilot Design Impact

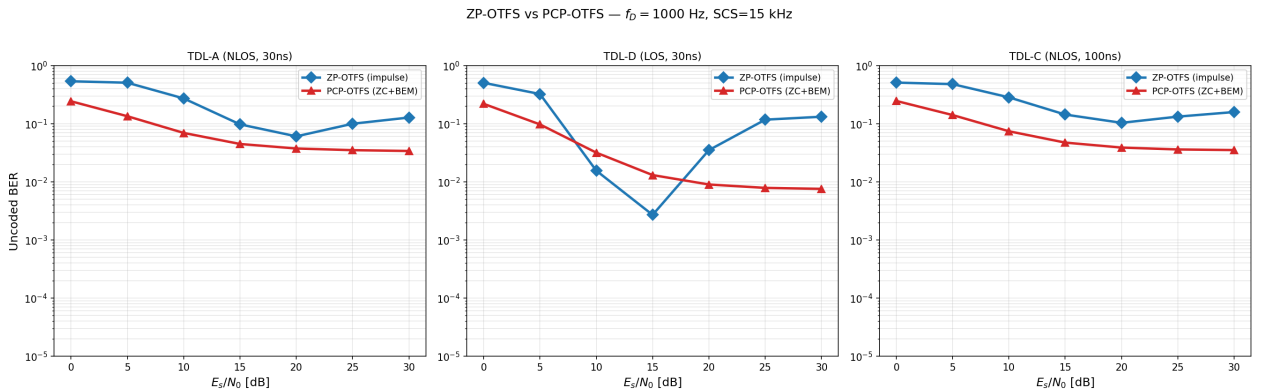


Figure 12: Impact of pilot design at SCS = 15kHz, $f_D = 1000$ Hz. PCP-OTFS (ZC pilot, red) consistently outperforms ZP-OTFS (impulse pilot, blue) by 1.5–4 $\times$.

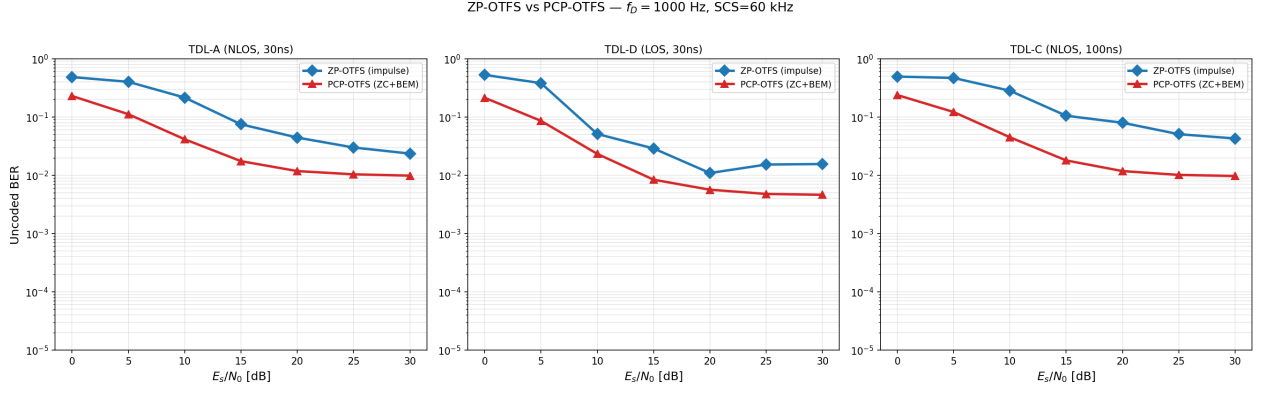


Figure 13: Impact of pilot design at SCS = 60 kHz, $f_D = 1000$ Hz. PCP improvement is even larger: 4–7 $\times$ across all channels.

10.3 Impact of SCS

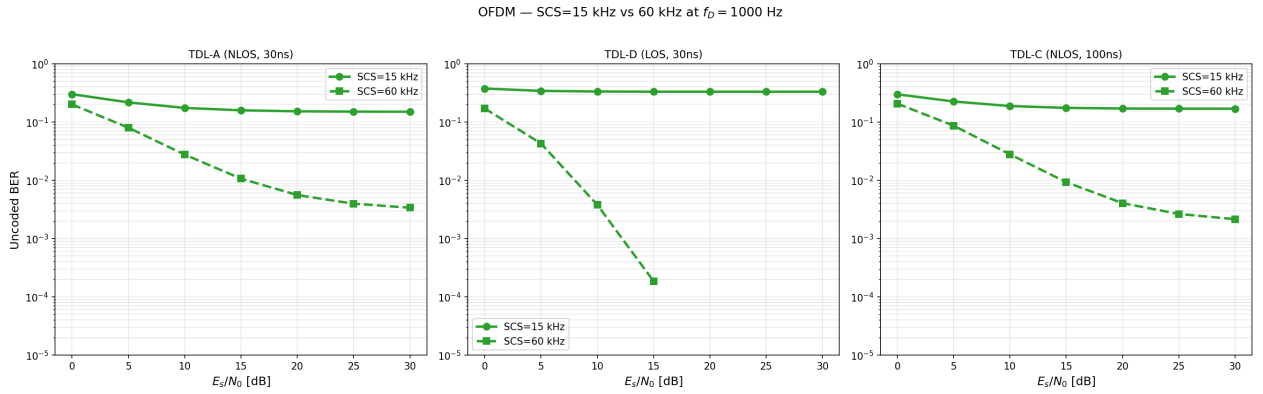


Figure 14: SCS impact on OFDM at $f_D = 1000$ Hz. SCS = 15 kHz (solid) vs 60 kHz (dashed). Higher SCS dramatically reduces ICI: BER drops from 15% to 0.6% in TDL-A.

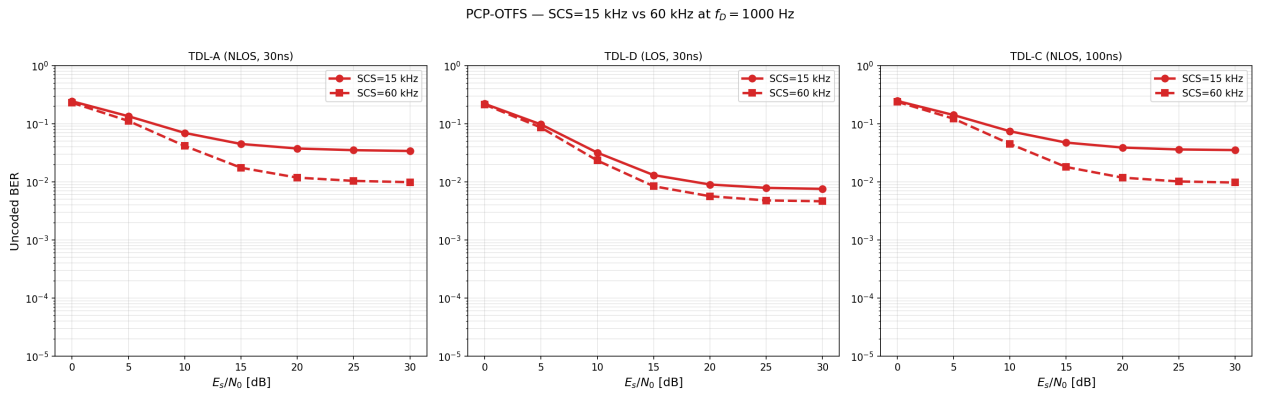


Figure 15: SCS impact on PCP-OTFS at $f_D = 1000$ Hz. Unlike OFDM, PCP-OTFS is largely SCS-independent: BER changes by less than 2 $\times$ between SCS = 15 and 60 kHz.

10.4 Summary of Winners

Table 5: Winning method at SNR = 20 dB. **O** = OFDM, **P** = PCP-OTFS, = = tie (both $< 10^{-4}$).

SCS	f_D	TDL-A			TDL-D			TDL-C		
		0	100	1k	0	100	1k	0	100	1k
15k		=	O	P	=	=	P	P	P	P
30k		P	P	O	=	=	O	P	P	O
60k		P	P	O	=	=	O	P	P	O

Pattern: PCP-OTFS wins at low Doppler (*DD diversity gain*) and at low SCS / high Doppler (*OFDM ICI vulnerability*). OFDM wins at high SCS / high Doppler (*short symbols minimize ICI*).

11 Conclusions

1. **Prior art oversimplifies OTFS evaluation.** Under practical conditions—estimated CSI, estimated noise, realistic pilot overhead—the OTFS advantage is more nuanced than commonly reported. The impulse pilot (ZP-OTFS) consistently underperforms both OFDM and PCP-OTFS.
2. **SCS is the dominant factor, not the waveform.** The normalized Doppler $f_D/\Delta f$ determines OFDM’s ICI penalty. At SCS = 60 kHz, OFDM handles $f_D = 1000$ Hz with $< 1\%$ BER; at SCS = 15 kHz, the same Doppler causes 15–33% BER. PCP-OTFS is largely SCS-independent.
3. **PCP-OTFS is the practical OTFS architecture.** The ZC pilot with GCE-BEM provides: (a) 14 dB PAPR reduction; (b) delay-spread-independent overhead; (c) per-subsymbol channel tracking via BEM.
4. **Channel estimation is the bottleneck.** ZP-OTFS with true CSI achieves BER = 0 at all Dopplers (SNR ≥ 10 dB). The waveform diversity gain is real but hidden by the $N = 14$ Doppler resolution limit.
5. **OTFS is most compelling for sub-6 GHz high-mobility:** Low SCS scenarios (FR1 coverage bands, high-speed rail, NTN with LEO satellites) where OFDM’s ICI is the performance limiter.

References

- [1] R. Hadani et al., “Orthogonal time frequency space modulation,” in *Proc. IEEE WCNC*, Mar. 2017.
- [2] P. Raviteja, K. T. Phan, Y. Hong, and E. Viterbo, “Embedded pilot-aided channel estimation for OTFS in delay-Doppler channels,” *IEEE Trans. Veh. Technol.*, vol. 68, no. 5, pp. 4906–4917, May 2019.
- [3] S. Sanoopkumar and A. Farhang, “A practical pilot for channel estimation of OTFS,” in *Proc. IEEE*, 2023.
- [4] C. Schripps, L. Torres-Figueroa, U. Monich, and H. Boche, “Channel estimation for OTFS using pilots with cyclic prefix in fractional Doppler channels,” in *Proc. IEEE ICC Workshops*, Jun. 2025.
- [5] R. Zedka et al., “Unique word channel estimation for oversampled OTFS,” arXiv:2601.09364, Jan. 2026.

- [6] Y. Liu et al., “Cross-domain OTFS detection via delay-Doppler decoupling,” *Entropy*, 2025.
- [7] D. C. Chu, “Polyphase codes with good periodic correlation properties,” *IEEE Trans. Inform. Theory*, vol. 18, no. 4, pp. 531–532, Jul. 1972.
- [8] G. B. Giannakis and C. Tepedelenlioğlu, “Basis expansion models and diversity techniques for blind identification and equalization of time-varying channels,” *Proc. IEEE*, vol. 86, no. 10, pp. 1969–1986, Oct. 1998.
- [9] S. Tomasin, A. Gorokhov, H. Yang, and J.-P. Linnartz, “Iterative interference cancellation and channel estimation for mobile OFDM,” *IEEE Trans. Wireless Commun.*, vol. 4, no. 1, pp. 238–245, Jan. 2005.
- [10] L. Rugini, P. Banelli, and G. Leus, “Low-complexity banded equalizers for OFDM systems in Doppler spread channels,” *EURASIP J. Adv. Signal Process.*, 2006.
- [11] R. L. Frank, S. A. Zadoff, and R. Heimiller, “Phase shift pulse codes with good periodic correlation properties,” *IRE Trans. Inform. Theory*, vol. 8, no. 6, pp. 381–382, Oct. 1962.